Fiscal Year:	FY 2015	**Task Last Updated:**	FY 03/20/2015
PI Name:	Harrison, Lynn Harrison		
Project Title:	The Effect of Microgravity on Mycobacteria		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	lynn.clary@lsuhs.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	318-675-7013
Organization Name:	Louisiana State University		
PI Address 1:	Pleasant Hall, 1146		
PI Address 2:			
PI Web Page:			
City:	Baton Rouge	**State:**	LA
Zip Code:	70802	**Congressional District:**	5
Comments:			
Project Type:	Ground,New Investigation	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	11/30/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 11/30/2014 per A. Chu/ARC and PI; original end date was 8/31/2014 (Ed., 7/14/14)		
Key Personnel Changes/Previous PI:	Final report: Camille Abshire - was a technician after Reneau Youngblood left the lab		
COI Name (Institution):	Ennis, Don (University of Louisiana, Lafayette)		
Grant/Contract No.:	NNX13AM02G		
Performance Goal No.:			
Performance Goal Text:			
	Mycobacteria are bacteria that are found in water, vegetation, dust, and soil, and certain types such as Mycobacterium tuberculosis, live and replicate only in cells that work to protect an animal or human. These immune cells are called macrophage, and M. tuberculosis is the cause of tuberculosis in humans. Mycobacterium avium and Mycobacterium marinum can also cause human disease as they too can live and survive inside macrophage. M. marinum causes a skin infection, where as M. avium has been linked with pulmonary infections. M. marinum and M. avium are two of the types of mycobacterium species that live in water. M. avium has been found in drinking water even after extensive chemical treatment of water systems, and was identified in water samples taken from the Russian space station Mir. All of these types of mycobacterium form biofilms and biofilms have been found on surfaces in the space station. A biofilm is formed when microorganisms assemble together and are enclosed by a protective film resulting in resistance to		

Task Description:	disinfecting agents or drug treatments. Since it is more difficult to kill bacteria in biofilms, human mycobacterium infections could be a problem during extended space flight. Human mycobacterium infections from M. avium, M. marinum, and even M. tuberculosis are more of a problem when the body is less able to fight infection, as in the case of AIDS patients on Earth. This is also an issue for space travel as astronauts have been found to have compromised immune systems. Studies have also demonstrated increased growth, thicker biofilm formation as well as an increase in virulence of other types of bacteria grown under microgravity to model space conditions. The aim of this work is to determine whether mycobacteria have enhanced virulence during space travel and what mechanisms they use to adapt to microgravity. We will use M. marinum as a model mycobacterium species, as its optimal growth is at 30-33°C and it is less problematic in terms of causing serious human infections. M. marinum also infects fish, and this provides a future small animal model for both ground-based studies and possible experiments on the International Space Station. We therefore hypothesize that M. marinum grown under conditions of microgravity will be more infectious in terms of entry and survival inside macrophage cells, and that this will be achieved by altering gene expression to promote survival under stressful conditions. Oxygen has been found to play a role in altering gene expression in Pseudomonas aeruginosa grown during space flight, and Salmonella enterica are more resistance to oxidative stress when grown under microgravity. We hypothesize that genes involved in surviving oxidative stress will be induced in M. marinum grown under microgravity. M. marinum grown under microgravity and normal gravity will be tested for their ability to infect and survive in macrophage, for resistance to hydrogen peroxide and for gene expression changes stimulated by microgravity. Specific mutant strains defective in detoxification processes, damage repair, or biofilm formation will also be generated and if time allows will also be tested for survival in macrophage after growth under microgravity. This one year proposal will therefore build on existing knowledge and will be used to develop a three year space biology grant. This will further extend our understanding of mycobacteria pathogenicity during space travel and potentially reveal mechanisms to target mycobacterium infections in space and on Earth. The proposed work is therefore consistent with the objectives of the NASA Space Biology Program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Mycobacterium marinum is a waterborne pathogenic mycobacteria. It can infect humans on the skin, but is a good model for other human pathogenic mycobacteria (Mycobacterium avium and Mycobacterium tuberculosis) that result in serious human lung infections. This work examined the stress response of M. marinum when grown under microgravity. By comparing the changes in gene expression this work suggests that the stress responses turned on during growth under simulated microgravity are similar to those found in M. tuberculosis inside human immune cells. The data suggests that a protein called SigH may be involved in the alteration of the stress response genes. This work therefore suggests that if we can identify a way of preventing growth under simulated microgravity we may be able to apply this knowledge to identifying drugs to prevent survival of pathogenic mycobacteria in humans on Earth and aid in the treatment of lung infections caused by pathogenic mycobacteria.
Task Progress:	The objective of this proposal was to determine whether waterborne pathogenic mycobacteria alter gene expression, become resistant to stress, and are more infectious when grown under low shear simulated microgravity (MG). Mycobacterium marinum was the model organism used in this study. This is a bacterium that naturally infects fish and amphibians to produce a tuberculosis-like disease in a variety of organs. In humans, M. marinum can cause a disease known as swimming pool granuloma, which is restricted to the skin due to the lower temperature of body extremities. The pathology is similar to dermal Mycobacterium tuberculosis infection and the infection route is through cuts or abrasions. Infection is usually limited to the skin in humans since M. marinum grows at 28-35°C. M. marinum infects and survives inside immune cells called macrophage by the same mechanisms as M. tuberculosis and Mycobacterium avium. M. tuberculosis and M. avium cause serious lung infections in humans. M. marinum is therefore a good model to understand how these pathogenic mycobacteria survive and adapt to the hostile environment inside macrophage. Since M. marinum and M. avium exist in the environment on Earth and can form biofilms in water systems, it is important to understand the effect of microgravity on these pathogenic mycobacteria as they could exist in the water systems on space vehicles. M. marinum and LHM4 (a derivative of M. marinum carrying a plasmid encoding kanamycin resistance and red fluorescent protein) were grown in high aspect ratio vessels (HARV) in a rotary cell culture system (RCCS) under normal gravity (NG) or MG. By following the optical density of the culture as well as the colony forming units per milliliter we determined that the growth of both strains was altered by growth under MG once the culture was in exponential phase. There was no difference in the growth for ~40 hrs (early exponential phase) but the growth of the MG culture was significantly slower after 40 hrs. The doubling time during early exponential was 6 hrs and after 40 hrs this increased to ~8 hrs for the NG culture and to ~15 hrs for the MG culture. We hypothesized that the slowing of growth may have been due to an oxidative stress in the bacteria. Hydrogen peroxide can cross membranes and so we repeated the growth studies adding catalase to the culture. Any hydrogen peroxide exiting the bacteria and existing in the medium would be detoxified by the catalase. This did not however improve the growth of the M. marinum under MG. To determine the effect of MG on the stress responses activated by the growth conditions, we used RNAseq to examine what genes were expressed. For RNAseq, the bacteria are harvested, RNA isolated and converted DNA (cDNA), and the cDNA sequenced. Using bioinformatics, the amount of expression of the different M. marinum genes were compared between the NG and MG samples. To make sure that we were examining only gene expression changes due to MG, only bacteria in early exponential growth were used in the RNAseq studies. Triplicate NG and MG cultures were used to generate samples of bacteria grown for ~40 hrs. We also grew triplicate cultures for 4 days and then diluted them again and grew them for another ~40 hrs so we could examine gene expression from bacteria exposed for a longer time. From these studies it was obvious that stress response pathways were activated in the MG cultures and the stress responses were at a higher expression level the longer they were exposed to MG. Expression seemed to change to be similar to responses of M. tuberculosis inside macrophage. This suggests that the MG stress was similar to that of mycobacteria inside immune cells. Using the MG growth conditions may be useful in the future to test drugs designed to kill mycobacteria during infection. We are in the process of identifying regulator proteins that may be involved in aiding mycobacteria survival under MG. Published studies indicate that other species of bacteria show altered resistance to hydrogen peroxide and acid (pH 3.5) once grown under MG. We tested our LHM4 strain and determined that as found for other bacteria M. marinum are more sensitive to killing by hydrogen peroxide but do not alter their resistance to acid conditions once grown under MG. M. marinum sensitivity to gamma radiation did not change also. This type of study may aid in designing disinfection

	techniques for future space travel. Since this was a 1 year grant there was limited time to test the survival capabilities of the M. marinum inside immune cells. An initial study did suggest that the M. marinum grown under MG have a growth advantage inside the immune cells during the first 2 days. More work is required to confirm these results. In summary, this study determined that waterborne mycobacteria alter their growth, expression of stress responses, and their sensitivity to oxidizing conditions when subjected to growth under MG. We can now use these results to identify important proteins required for adaptation to growth and survival under MG and this may aid in the design of new disinfection procedures and drugs to prevent human infection during space travel.
Bibliography Type:	Description: (Last Updated: 02/27/2023)
Abstracts for Journals and Proceedings	Abshire C, Ennis DG, Concha M, Baddoo M, Flemington E, Scott R, Harrison L. "The effects of simulated microgravity on Mycobacterium marinum: growth, survival, and gene expression." 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. 30th Annual Meeting of the American Society for Gravitational and Space Research, Pasadena, CA, October 22-26, 2014. , Oct-2014
Articles in Peer-reviewed Journals	Abshire CF, Prasai K, Soto I, Shi R, Concha M, Baddoo M, Flemington EK, Ennis DG, Scott RS, Harrison L. "Exposure of Mycobacterium marinum to low-shear modeled microgravity: effect on growth, the transcriptome and survival under stress." npj Microgravity. 2016 Dec 1;2:16038. eCollection 2016. https://doi.org/10.1038/npjmgrav.2016.38 ; PubMed PMID: 28725743; PubMed Central PMCID: PMC5515531 , Dec-2016
Significant Media Coverage	LSU Health Shreveport. "Cover story about PI Lynn Harrison receiving NASA grant to study The Effect of Microgravity on Mycobacteria." LSU Health Shreveport. 2013 Nov-Dec;13(6)., Dec-2013
Significant Media Coverage	Sims C. "PI Harrison interviewed about her NASA research in article, 'NASA funds local zero gravity and bacteria research.' " KTBS station, Shreveport, LA. February 28, 2014. http://www.ktbs.com/story/24856873/2014/02/28/nasa-funds-local-zero-gravity-and-bacteria-research ; accessed 3/24/15., Feb-2014